

DBMS Assignment: Online Food Delivery System

Database Schema Specification & System Architecture Documentation

1. Name and Purpose of the Application

Application Name: Online Food Delivery System

Purpose:

An online food delivery system enables customers to browse partner restaurants, view menu offerings, place orders, process payments, and track order fulfillment. The relational database accurately models and stores interactions between customers, restaurants, food menus, orders, financial payments, and delivery personnel.

2. Table Specifications & Data Dictionary

The application schema consists of 8 core tables designed to maintain data integrity and eliminate redundancy.

Table 1: Customer

Stores profile information and authentication credentials for platform users.

Column	Data Type	Key	Description
Customer_ID	INT ▾	PK	Unique ID of customer
Name	VARCHAR(100) ▾		Full name of customer
Email	VARCHAR(100) ▾	UNIQUE	Contact email address

Column	Data Type	Key	Description
Phone	VARCHAR(15) ▾	UNIQUE	Primary phone number
Password	VARCHAR(255) ▾		Account login password

Table 2: Restaurant

Stores profiles of food establishments onboarded to the system.

Column	Data Type	Key	Description
Restaurant_ID	INT	PK	Unique restaurant ID
Restaurant_Name	VARCHAR(100)		Name of restaurant
Phone	VARCHAR(15)		Restaurant contact number
Address	VARCHAR(255)		Restaurant physical location
Rating	DECIMAL(2,1)		Average rating score

Table 3: Menu_Item

Stores individual catalog items offered by each restaurant.

Column	Data Type	Key	Description
Item_ID	INT ▾	PK	Unique food item ID
Restaurant_ID	INT ▾	FK	Restaurant offering the item
Item_Name	VARCHAR(100) ▾		Name of food item

Column	Data Type	Key	Description
Description	VARCHAR(255) ▾		Brief item description
Price	DECIMAL(10,2) ▾		Unit price of item
Availability	BOOLEAN ▾		Availability flag

Relationship Note: One restaurant can offer many menu items (1 : Many).

Table 4: Orders

Stores summary records of transactions initiated by customers.

Column	Data Type	Key	Description
Order_ID	INT ▾	PK	Unique order ID
Customer_ID	INT ▾	FK	Customer placing the order
Restaurant_ID	INT ▾	FK	Restaurant receiving order
Order_Date	DATETIME ▾		Timestamp of placement
Total_Amount	DECIMAL(10,2) ▾		Total transaction value
Order_Status	VARCHAR(30) ▾		Current status of order

Example Statuses: Placed, Preparing, Out for Delivery, Delivered, Cancelled.

Table 5: Order_Item

Associates specific menu items and quantities with a placed order.

Column	Data Type	Key	Description
Order_Item_ID	INT ▾	PK	Unique order-item record ID
Order_ID	INT ▾	FK	Related order ID
Item_ID	INT ▾	FK	Ordered menu item ID
Quantity	INT ▾		Quantity ordered
Price	DECIMAL(10,2) ▾		Unit price at time of order

Design Requirement: Serves as a junction table resolving the Many-to-Many relationship between **Orders** and **Menu_Item**.

Table 6: Delivery_Partner

Stores records of personnel responsible for fulfilling order deliveries.

Column	Data Type	Key	Description
Delivery_ID	INT	PK	Unique delivery partner ID
Name	VARCHAR(100)		Delivery partner full name
Phone	VARCHAR(15)	UNIQUE	Contact phone number
Vehicle_Number	VARCHAR(20)	UNIQUE	Vehicle registration identifier
Status	VARCHAR(30)		Operational availability status

Table 7: Payment

Tracks financial transaction details attached to each order.

Column	Data Type	Key	Description
Payment_ID	INT ▾	PK	Unique payment record ID
Order_ID	INT ▾	FK	Related order ID
Payment_Date	DATETIME ▾		Timestamp of transaction
Amount	DECIMAL(10,2) ▾		Total amount billed
Payment_Method	VARCHAR(30) ▾		Selected payment channel
Payment_Status	VARCHAR(30) ▾		Transaction status

Example Payment Methods: UPI, Card, Cash.

Table 8: Address

Stores saved delivery physical locations linked to customer accounts.

Column	Data Type	Key	Description
Address_ID	INT ▾	PK	Unique address ID
Customer_ID	INT ▾	FK	Owner customer ID
House_No	VARCHAR(50) ▾		House / Apartment details
Street	VARCHAR(100) ▾		Street name

Column	Data Type	Key	Description
City	VARCHAR(50) ▾		City name
Pincode	VARCHAR(10) ▾		Postal code

3. Relationships Between Tables

Logical Mapping Overview

- **Customer**
 - 1 : Many → **Address**
 - 1 : Many → **Orders**
- **Restaurant**
 - 1 : Many → **Menu_Item**
 - 1 : Many → **Orders**
- **Orders**
 - 1 : Many → **Order_Item**
 - 1 : 1 → **Payment**
- **Menu_Item**
 - 1 : Many → **Order_Item**
- **Delivery_Partner**
 - 1 : Many → **Orders**

Detailed Cardinality Rules

Primary Entity	Target Entity	Cardinality	Functional Explanation
Customer ▾	Address ▾	1 : Many ▾	A customer can save multiple delivery addresses
Customer ▾	Orders ▾	1 : Many ▾	A customer can place multiple historical orders

Primary Entity	Target Entity	Cardinality	Functional Explanation
Restaurant ▾	Menu_Item ▾	1 : Many ▾	A restaurant manages multiple food catalog items
Restaurant ▾	Orders ▾	1 : Many ▾	A restaurant can receive multiple customer orders
Orders ▾	Order_Item ▾	1 : Many ▾	An order can contain multiple individual line items
Menu_Item ▾	Order_Item ▾	1 : Many ▾	A single menu item can appear across many orders
Orders ▾	Payment ▾	1 : 1 ▾	Each order possesses exactly one payment record
Delivery_Partner ▾	Orders ▾	1 : Many ▾	A partner can handle multiple fulfilled orders over time

4. Architectural System Overview

The database design initiates at the **Customer** entity, capturing user profiles. A customer maintains one-to-many relationships with both saved **Addresses** and active **Orders**.

Each **Restaurant** lists multiple **Menu_Items**. When an order is created, the **Orders** table logs master transactional details (time, status, total cost), while the **Order_Item** table breaks down specific quantities and locked-in historical pricing per item.

Financial tracking is isolated within the **Payment** table (1:1 mapping with Orders), and physical fulfillment is delegated through the **Delivery_Partner** table. This structured normalization avoids duplicate data entries and guarantees integrity via Foreign Key enforcement.

5. Schema Summary List

No.	Table Name	Core Operational Purpose
1	Customer	Account management and customer details
2	Restaurant	Merchant profiles and physical locations
3	Menu_Item	Product catalog linked to restaurants
4	Orders	Master transaction receipts and tracking
5	Order_Item	Itemized line items within each order
6	Delivery_Partner	Fulfillment staff and vehicle registration
7	Payment	Billing records and transaction status
8	Address	Customer delivery location profiles



Entity-Relationship
(ER) DIAGRAM



